

PROJECT DESIGN PHASE: SOLUTION ARCHITECTURE

Date: 15 February 2025

Team ID: SWTID-2026-9692

Project Name: AI WeatherWise API

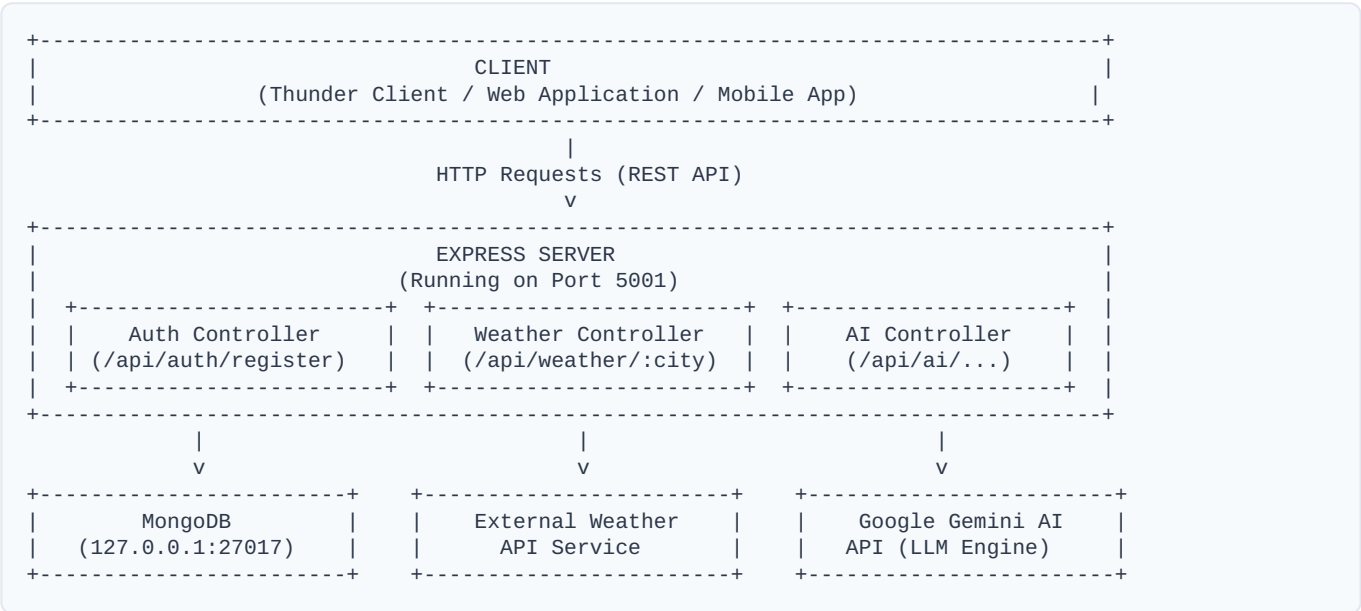
Maximum Marks: 4 Marks

1. Solution Architecture Overview

Solution architecture bridges business requirements and tech implementation. Its main goals are:

- Identify optimal technology stack and patterns to address real-world weather planning challenges.
- Provide clear specification of software layers, routes, and authentication flow for stakeholders.
- Define clear backend API contracts and third-party service dependencies.

2. Architecture Diagram & System Flow



3. Key System Components

- **Express.js REST API (server.js):** Operates on port 5001 to handle routes for authentication, weather fetching, and AI interaction.
- **MongoDB (127.0.0.1:27017):** Local database instance storing user registration profiles and securely hashed credentials.
- **JWT Security Layer:** Issues signed authorization tokens upon account registration (`/api/auth/register`) and sign-in (`/api/auth/login`).
- **Weather Integration Endpoint:** Retrieves live meteorological data including city temperature, humidity, wind speeds, and condition codes.
- **Google Gemini AI Integration:** Synthesizes live weather metrics into human-readable summaries (`/api/ai/weather-summary`) and tailored outfit tips (`/api/ai/weather-recommendation`).